

Kanav Nagpal

Computer Engineering | Data Science, Web Development, AI, and Hardware
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EDUCATION

University of Massachusetts Amherst, Amherst, MA

Anticipated Graduation: May 2029

B.S. with Honors in Computer Engineering, Minor in Mathematics | Commonwealth Honors College | GPA: 4.0

Relevant coursework: Data Structures and Algorithms (in progress), Introduction to Digital and Computer Systems (ECE 124), Circuits and Electronics Honors Colloquium (ECE H210, in progress), Honors 201H: Ideas That Change the World (in progress), Python programming, Calculus I

Shiv Nadar School, Faridabad, Haryana, India

Aug 2017 – May 2025

High School Diploma | GPA: 4.0

RESEARCH EXPERIENCE

Undergraduate Researcher, Formal Verification Research Group

Sep 2026 – Present

UMass Amherst | Faculty mentor: Professor Ciesielski

- Participate in a self-directed, faculty-mentored research group studying formal verification of arithmetic circuits
- Present weekly findings to the professor and two other undergraduate researchers for feedback

Undergraduate Research Assistant, High-Dimensional Signal Processing Group

Jun 2026 – Present

UMass Amherst | Mentors: Professor Marco Duarte and researcher Ritik Shah

- Run Python experiments on hyperspectral image super-resolution algorithms to measure robustness under varying conditions, using the Unity HPC cluster through SLURM
- Extend published research toward a journal submission on standardized benchmarking in remote sensing

TEACHING AND MENTORING

Undergraduate Instructional Assistant, ECE 124: Introduction to Digital and Computer Systems

Sep 2026 – Present

UMass Amherst | Instructor: Professor Ciesielski

- Hold weekly office hours and grade assignments and exams for the course

Peer Tutor (Informal)

Apr 2023 – Present

- Provide one-on-one tutoring in math and science to students and peers
- Break complex STEM concepts into step-by-step explanations and adjust my approach to each student's needs and pace

PROJECTS

Interlace: Hindi-English Code-Switched Translation

Oct 2026 – Present

Python, PyTorch | github.com/kn5225/interlace

- Implementing a Transformer for Hindi-English code-switched translation from scratch in PyTorch, with no pretrained translation model
- Built scaled dot-product attention with boolean masking and a single-head attention module, validated against PyTorch's reference implementation and hand-worked examples, with tests confirming no future-token leakage

HandCast: Gesture-Controlled Interface

Jul 2026

C++, Arduino (ATmega328P), HC-SR04, Python, PySerial, AutoHotkey v2 | github.com/kn5225/handcast

- Built an embedded pipeline in C++ that converts ultrasonic distance readings into smoothed data using an on-board exponential moving average filter, streamed to a computer over UART/USB
- Developed two host-side modes: a background AutoHotkey v2 service that sends system-wide keystrokes for hands-free media, document, and slide control, and a native Python SDK for gesture-based menu navigation in custom game loops
- Organized the repository as a modular monorepo that keeps firmware, OS-level drivers, and SDK integrations in sync

Simmr: Recipe and Meal Planning Platform

Apr 2026 – Present

React, Vite, Node.js, Express, Spoonacular API, Stripe | github.com/kn5225/Simmr

- Building a full-stack recipe app with recipe search, a Stripe-powered ingredient checkout (test mode), a step-by-step cooking mode with automatic timers, and a pantry tracker; deploying through Vercel and Railway
- Writing all code by hand, using AI only for conceptual guidance and code review, to learn full-stack architecture

PrereqCheq: UMass Prerequisite Checker

Jan 2026 – May 2026

HTML, CSS, JavaScript, Supabase (Data API, Cron, Auth) | github.com/kn5225/prereqcheq

- Built an academic planning tool that queries relational course data to resolve prerequisites and suggest registration paths across 80+ UMass courses
- Implemented a public interface for crowdsourced course corrections and an admin dashboard to review, approve, and manage database updates; deployed on GitHub Pages

PokeCLI: Command-Line Pokémon Battle Simulator

Jan 2025 – Mar 2025

Python, AutoHotkey v2 | github.com/kn5225/pokecli

- Built a turn-based battle simulator with type-effectiveness logic and move PP tracking, later extended with AutoHotkey v2 hotkey automation

Sortwise: AI Waste Segregation System

Apr 2024 – Jan 2025

Python, OpenCV, Keras, TensorFlow, NumPy | github.com/kn5225/Sortwise

- Led the programming for a real-time computer vision system that classifies 8 waste categories, designed for integration into a self-sorting waste bin
- Trained a custom image classification model on 90+ images using Google's Teachable Machine, reaching 94% accuracy; coordinated a team through data collection, labeling, and testing

TECHNICAL TEAMS AND LEADERSHIP

Team Member, AntWeight BattleBots Competition 2026, Team 6, UMass Amherst

Sep 2025 – Apr 2026

- Contributed design input on a 3D-printed, CAD-modeled chassis for a combat robot built around a horizontal spinner weapon
- Designed for full reversibility with symmetric wheel placement so the robot could keep driving if flipped, and worked with an ESP32 microcontroller for onboard control; the team disbanded before firmware development and competition entry

Website Developer and Deputy Director of Operations, SNSF Model UN 2024

Jun 2024 – Aug 2024

Shiv Nadar School Faridabad

- Updated the conference website's content and branding for 150+ delegates, adding articles and committee images, updating committee listings, and aligning the color scheme with school branding
- Also served as Deputy Director of Operations, coordinating conference logistics and acting as de facto lead in the Director's absence

Founding Member, Technology Council

Aug 2023 – Jan 2025

Shiv Nadar School Faridabad

- Co-founded the school's first student-led Technology Council, focused on digital literacy
- Led a cybersecurity workshop for 200+ students in Grades 3 to 6 and an AI tools workshop for teachers; wrote questions for and helped host the annual STEM quiz in 2023 and 2024

HONORS, AWARDS, AND CERTIFICATIONS

- Commonwealth Honors College, UMass Amherst: Sep 2025 – Present
- Dean's List, UMass Amherst: 2×, 2025–2026
- Sundar Pichai Award in Computer Science (top CS student, school-wide): 2024, 2025
- Ramanujan Award in Mathematics (top math student, school-wide, 7 consecutive years): 2019–2025
- Distinction, Euclid Contest, University of Waterloo: 2024
- IBM Web Development Fundamentals certification: 2026

SKILLS

Programming languages: Python, C++, MATLAB, Verilog, Bash, JavaScript, HTML/CSS

Machine learning and data: PyTorch (basic), OpenCV, Keras, TensorFlow, scikit-learn, NumPy, SciPy, pandas, matplotlib

Web development: React, Node.js, Express, Vite, Supabase, MySQL/PostgreSQL

Hardware and systems: Arduino, ESP32, HC-SR04 sensors, Linux, SLURM, AutoHotkey

Tools: Git/GitHub, Microsoft Office, Google Workspace

Languages: English, Hindi

Last updated October 2026